Sinan Emre

Researcher in Mechanical Design of Legged Robots and Actuators

Genova, Italy | +39 342 775 5998 | sinanemre94@gmail.com | sinan.emre@iit.it
linkedin.com/in/sinan-emre-04888694 | scholar.google.com/citations?user=wjPPl1AAAAAJ
Available from February 2027. Full-time or contract, on-site or remote.

Experience

Istituto Italiano di Tecnologia (IIT), Dynamic Legged Systems lab, Genova, Italy Feb 2023 – present

PhD Researcher in Bioengineering and Robotics Nov 2023 – present

Research Fellow in Mechanical Engineering Feb 2023 – Nov 2023

- **Carpal-claw mechanism:** carried out the mechanical design of a mechanism that helps quadruped robots negotiate high obstacles, and tested it on Unitree Aliengo (ICRA 2024).
- **Load-based variable transmission:** passive knee mechanism (pre-tensioned torsional springs and a four-bar linkage) that increases the knee transmission ratio under load, by up to 40% in simulation; built a 3D-printed prototype (first author, ICAR 2025).
- **Compact hydraulic power unit:** simulated and experimentally evaluated the unit, and integrated it on the HyQReal2 hydraulic quadruped (first author, AIM 2026).
- **Stair-climbing study:** built the simulations for a study of the optimal body height for quadrupeds walking on stairs (CLAWAR 2023).
- **HyQReal and Unitree robots:** system integration, experiments, mechanical maintenance and public demos on HyQReal, including work in the INAIL project; experiments on Unitree Aliengo; demos on Unitree Go2.

Özyeğin University, Biomechatronics Lab, Istanbul, Turkey Jun 2018 – Nov 2022

Research and Design Engineer Apr 2021 – Nov 2022

- Developed an additively manufactured actuator unit and quadruped prototype using composite materials; built FEM models of composite structures and studied how infill types behave under load.
- Carried out quality control of every component on the robot prototypes, and experiments on actuator units and quadruped prototypes.

Research Engineer, CAD Design and Optimization Apr 2020 – Apr 2021

- **Compact quadruped actuator:** sole designer; 48 Nm continuous torque at 1.6–1.7 kg (first author, ACTUATOR 2022).
- **12-DOF quadruped:** sole designer of a topology-optimized quadruped driven by these actuators.
- Built parametric FEM models in Ansys, selected materials and ran topology optimization for critical load cases; built dynamic simulation models of floating-base robots; handled manufacturing follow-up and quality control of mechanical parts.

Lab and Teaching Assistant; earlier Lab Assistant, Mechanical Design Jun 2018 – Apr 2020

- CoEx lower-body exoskeleton (TÜBİTAK project), from its start: designed the limbs, verified the series elastic actuator, ran FEM on mechanical parts and tracked manufacturing; designed mechanism parts for pneumatic muscles.
- Topology optimization of the torsional spring in a series elastic actuator, with design verification.

Engineering internships, Istanbul, Turkey 2015 – 2017

Anadolu Isuzu, manufacturing engineering (2017); Çiftel Elektromekanik, quality control (2016); Özdemir Vargel, CNC operation (2015).

Skills

Robot platforms: HyQReal2 hydraulic quadruped, Unitree Aliengo and Go2, own 12-DOF quadruped prototypes, CoEx lower-body exoskeleton.

CAD and drawings: Siemens NX, CATIA, SolidWorks, Autodesk Inventor; engineering drawings with GD&T and tolerances; manufacturing follow-up with machine shops and suppliers.

Analysis: FEM/FEA (Ansys, Abaqus, ANSA-META), topology and design optimization, multibody dynamics (MSC Adams), MATLAB.

Manufacturing and experiments: additive manufacturing, composite materials, NI LabVIEW, robot and actuator experiments.

Robot software: ROS, ROS 2.

Education

PhD, Bioengineering and Robotics Nov 2023 – present
Istituto Italiano di Tecnologia / University of Genova, Italy
Thesis (working title): *Design of the Next Generation Quadruped Robots.*

MSc, Mechanical Engineering Sep 2019 – Aug 2022
Özyeğin University, Istanbul, Turkey. 100% merit scholarship, Özyeğin merit grant, TÜBİTAK grant.
Thesis: *Design, development, and real-life implementation of a high power quadruped robot.* Two prototypes of the quadruped built and tested in gait experiments.
Supervisors: Ramazan Ünal, Barkan Uğurlu.

BSc, Mechanical Engineering Sep 2013 – Aug 2018
Özyeğin University, Istanbul, Turkey. 50% merit scholarship.

Selected publications and patent

- **S. Emre**, M. Villa, V. Barasuol, C. Semini. Design and performance evaluation of a compact hydraulic power unit for legged robots. *IEEE/ASME AIM*, 2026.
- **S. Emre**, V. Barasuol, M. Villa, C. Semini. Load-based variable transmission mechanism for robotic applications. *IEEE ICAR*, 2025, pp. 73–79.
- V. Barasuol, **S. Emre**, V. S. Medeiros, A. Bratta, C. Semini. Introducing the carpal-claw: a mechanism to enhance high-obstacle negotiation for quadruped robots. *IEEE ICRA*, 2024, pp. 3457–3463.
- **S. Emre**, B. Akın, E. C. Özçınar, B. Balcı, P. Şendur, Ö. Bebek, R. Ünal, B. Uğurlu. Topology optimization-based design and development of a compact actuator with a high torque-to-weight ratio for quadrupeds. *ACTUATOR 2022*, VDE.
- M. C. Yıldırım, P. Şendur, A. T. Kansızoğlu, U. Uras, O. Bilgin, **S. Emre**, G. G. Yapıcı, M. Arık, B. Uğurlu. Design and development of a durable series elastic actuator with an optimized spring topology. *Proc. IMechE Part C*, 235(24), 2021.
- M. C. Yıldırım, B. Uğurlu, P. Şendur, **S. Emre**, M. Derman, S. Çoruk. Wearable lower extremity exoskeleton. *US Patent 12,521,298 B2*, 2026; *EP 3 986 353 B1*, 2024.

Full list (8 papers) on Google Scholar.

Languages and interests

Languages: Turkish (native) | English (fluent) | German (A2–B1) | Italian (A2)
Interests: Basketball, professional licensed player (2011–2013). Sailing, 1-star licence (2017). Wing Chun kung fu, self-defence instructor since 2022.